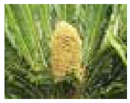
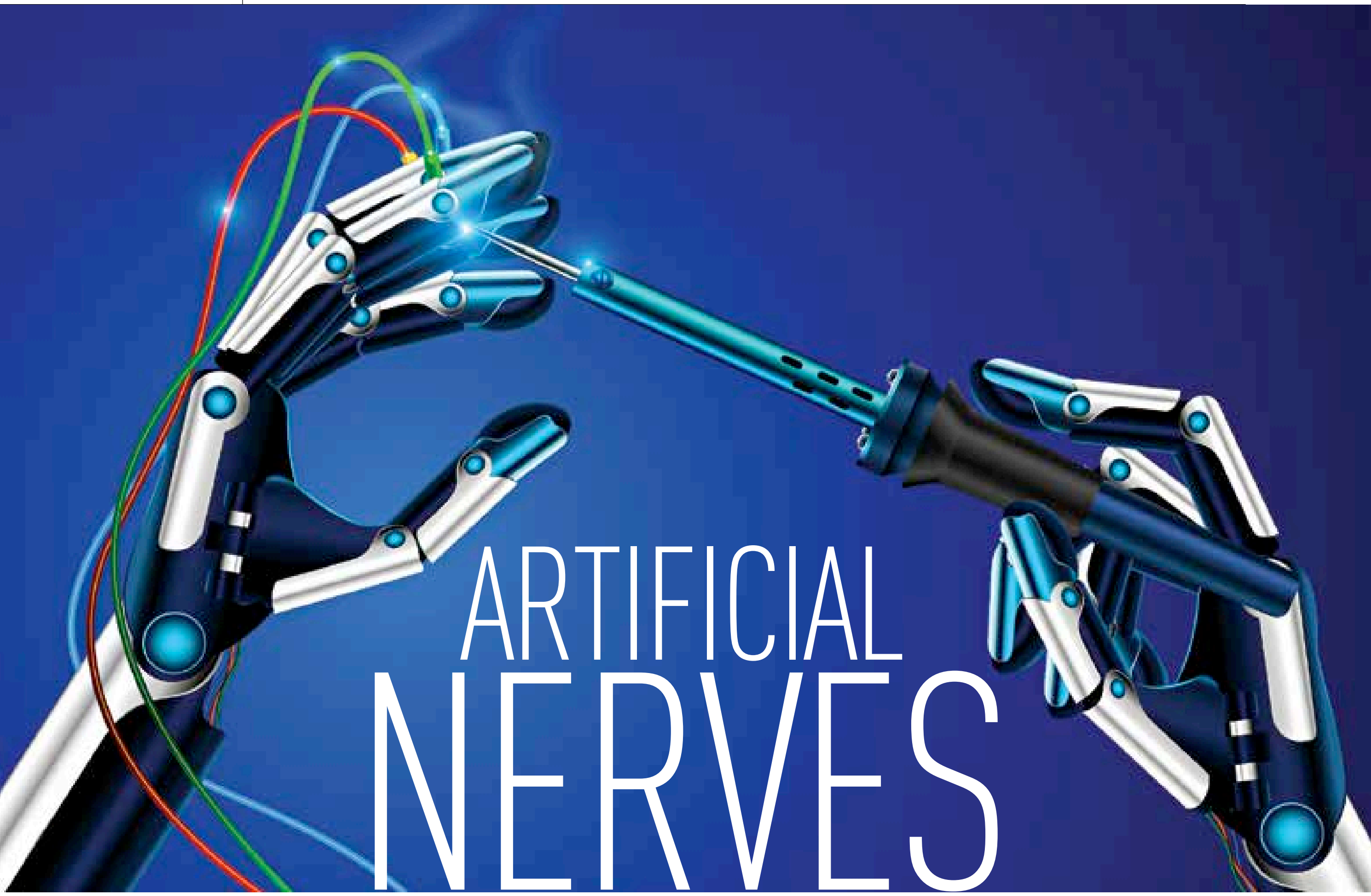
The science behind memories
Researchers have figured out what exactly causes memories to form and fade in the human mind. **https://dgit.in/sep19-05**



Prehistoric plant reproduction
Cycads, palm-like prehistoric trees are flowering and may reproduce for the first time in human history. **https://dgit.in/sep19-06**



ARTIFICIAL NERVES

An artificial nervous system that allows machines to feel can benefit humans as much as robots

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Human limbs provide a tremendous amount of continuous sensory feedback to the human brain. These include the sensation of touch, pressure, heat, orientation and pain. It is this information that allows humans to effectively handle a water balloon, take their fingers away from a shard of glass, feel a pin prick or reflexively pull away fingers from a hot surface such as a toaster or a stove. It is the nervous system, along with the muscles residing in the limbs that provide all these sensations. However, robots and those with prosthetic limbs, do not enjoy the benefits of being able to feel. Scientists in many universities around the world are coming up with innovative ways to address this problem.

All over the skin, are specialised nerve endings known as mechanoreceptors. When a force is applied on these receptors, they change their voltage. The changes in voltage from a number of mechanoreceptors are relayed by neurons. This feeling passes through the spinal cord to the human brain. The amount of pressure is determined by the frequency of the electric pulses, or hertz. A higher frequency indicates a higher amount of pressure. Artificial sensory nerves are designed along similar lines. Piezoelectric receptors convert mechanical stress into electricity. The more pressure is applied, the more electricity is generated. This electricity is passed on to ring oscillators, that generate electrical pulses based on the input voltage. A higher pressure on the receptors means a higher frequency of electric pulses, similar to the neurons in the human body. Another device known as the synaptic transistor converts the inputs from multiple ring oscillators. This artificial nervous system developed by researchers at Stanford and Seoul National University can one day be used to restore a sense of touch to amputees, and provide one to robots. The applications of this invention for soft, modular, medical, educational and evolutionary robotics are particularly promising.

One of the senior researchers on the project, Zhenan Bao says, “we take skin for granted but it’s a complex